



Lecture 3

- Gradient, Divergence, Curl
- In-class problems 1

Del operator



Open expression of ∇

Cartesian:

$$\nabla_{\text{red}} = \hat{x} \frac{\partial_{\text{red}}}{\partial x} + \hat{y} \frac{\partial_{\text{red}}}{\partial y} + \hat{z} \frac{\partial_{\text{red}}}{\partial z}$$

Cylindrical:

$$\nabla_{\text{red}} = \hat{r} \frac{\partial_{\text{red}}}{\partial r} + \hat{\phi} \frac{1}{r} \frac{\partial_{\text{red}}}{\partial \phi} + \hat{z} \frac{\partial_{\text{red}}}{\partial z}$$

Spherical:

$$\nabla_{\text{red}} = \hat{R} \frac{\partial_{\text{red}}}{\partial R} + \hat{\theta} \frac{1}{R} \frac{\partial_{\text{red}}}{\partial \theta} + \hat{\phi} \frac{1}{R \sin \theta} \frac{\partial_{\text{red}}}{\partial \phi}$$

Divergence of a vector field

in: vector out: scalar.

$$\vec{A} = \hat{x}A_x + \hat{y}A_y + \hat{z}A_z$$

$$\nabla \cdot \vec{A}$$
$$\text{div } \vec{A}$$

If $\text{div } \vec{A}$ is 0 everywhere, the field is called

Solenoidal field,
divergence-free
divergenceless

At a specific point in space.

$$\text{div } \vec{A} > 0,$$

$$\text{div } \vec{A} < 0,$$

$$\text{div } \vec{A} = 0.$$

the point contains a source of $\vec{A}$ field lines

Sink of $\vec{A}$ field lines

no source, no sink



Divergence of a vector field

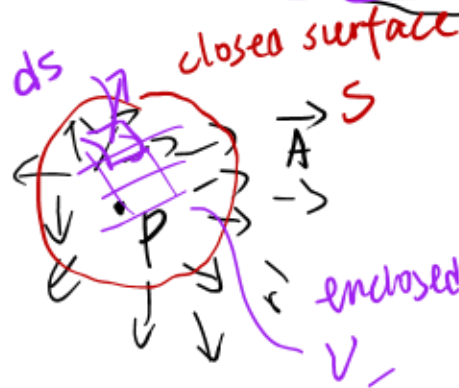


$$\nabla \cdot \vec{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}$$

in- vector out- scalar.

$$\nabla = \hat{x} \frac{\partial}{\partial x} + \hat{y} \frac{\partial}{\partial y} + \hat{z} \frac{\partial}{\partial z}$$
$$\vec{A} = \hat{x} A_x + \hat{y} A_y + \hat{z} A_z$$

Divergence is defined as the outward flux per unit volume in the limit of an infinitesimally small volume.


$$\lim_{V \rightarrow 0} \frac{\oint_S \vec{A} \cdot d\vec{s}}{V} \equiv \text{div } \vec{A}$$

Divergence of a vector field



Cartesian

$$\nabla \cdot \mathbf{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}$$

Cylindrical

$$\nabla \cdot \mathbf{A} = \frac{1}{r} \frac{\partial}{\partial r} (r A_r) + \frac{1}{r} \frac{\partial A_\phi}{\partial \phi} + \frac{\partial A_z}{\partial z}$$

Spherical

$$\begin{aligned} \nabla \cdot \mathbf{A} = & \frac{1}{R^2} \frac{\partial}{\partial R} (R^2 A_R) \\ & + \frac{1}{R \sin \theta} \frac{\partial}{\partial \theta} (A_\theta \sin \theta) \\ & + \frac{1}{R \sin \theta} \frac{\partial A_\phi}{\partial \phi} \end{aligned}$$

Curl of a vector field



Curl is defined as the circulation per unit area in the limit of an infinitesimally small area.


$$\lim_{S \rightarrow 0} \frac{\oint_C \vec{A} \cdot d\vec{l}}{S} \underbrace{\quad}_{\text{mag}} \underbrace{\hat{n}}_{\text{dir.}} \equiv \nabla \times \vec{A}$$

! use right-hand rule to find.



Curl of a vector field



Curl is defined as the circulation per unit area in the limit of an infinitesimally small area.

in vector

$$\nabla \times \vec{A} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix} = \hat{x} \left[\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \right] + \hat{y} \left[\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} \right] + \hat{z} \left[\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right]$$

out vector

$$\left\{ \begin{array}{lll} A_x & y & z \\ A_y & z & x \\ A_z & y & x \end{array} \right.$$

$$\vec{A} = \hat{x} x^2 \quad A_x = x^2 \quad \nabla \times \vec{A} = 0$$

★ Vector field with $\nabla \times \vec{A} = 0$ everywhere - field is conservative/irrotational!

Curl of a vector field



Cartesian

$$\begin{aligned}\nabla \times \mathbf{A} &= \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix} \\ &= \hat{\mathbf{x}} \left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \right) \\ &\quad + \hat{\mathbf{y}} \left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} \right) \\ &\quad + \hat{\mathbf{z}} \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right)\end{aligned}$$

Cylindrical

$$\begin{aligned}\nabla \times \mathbf{A} &= \frac{1}{r} \begin{vmatrix} \hat{\mathbf{r}} & \hat{\boldsymbol{\phi}} r & \hat{\mathbf{z}} \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \phi} & \frac{\partial}{\partial z} \\ A_r & rA_\phi & A_z \end{vmatrix} \\ &= \hat{\mathbf{r}} \left(\frac{1}{r} \frac{\partial A_z}{\partial \phi} - \frac{\partial A_\phi}{\partial z} \right) \\ &\quad + \hat{\boldsymbol{\phi}} \left(\frac{\partial A_r}{\partial z} - \frac{\partial A_z}{\partial r} \right) \\ &\quad + \hat{\mathbf{z}} \frac{1}{r} \left[\frac{\partial}{\partial r} (rA_\phi) - \frac{\partial A_r}{\partial \phi} \right]\end{aligned}$$

Spherical

$$\begin{aligned}\nabla \times \mathbf{A} &= \frac{1}{R^2 \sin \theta} \begin{vmatrix} \hat{\mathbf{R}} & \hat{\boldsymbol{\theta}} R & \hat{\boldsymbol{\phi}} R \sin \theta \\ \frac{\partial}{\partial R} & \frac{\partial}{\partial \theta} & \frac{\partial}{\partial \phi} \\ A_R & RA_\theta & (R \sin \theta) A_\phi \end{vmatrix} \\ &= \hat{\mathbf{R}} \frac{1}{R \sin \theta} \left[\frac{\partial}{\partial \theta} (A_\phi \sin \theta) - \frac{\partial A_\theta}{\partial \phi} \right] \\ &\quad + \hat{\boldsymbol{\theta}} \frac{1}{R} \left[\frac{1}{\sin \theta} \frac{\partial A_R}{\partial \phi} - \frac{\partial}{\partial R} (RA_\phi) \right] \\ &\quad + \hat{\boldsymbol{\phi}} \frac{1}{R} \left[\frac{\partial}{\partial R} (RA_\theta) - \frac{\partial A_R}{\partial \theta} \right]\end{aligned}$$

Problem 2:

In-class example

Examine homework problem 3.58 below. **Without doing any math**, which of these do you expect to be conservative? **Why?**

$$\text{h. } \mathbf{B} = \overset{A_x}{\hat{x}}x^2 - \overset{A_y}{\hat{y}}y^2 + \overset{A_z}{\hat{z}}2z$$

$$\nabla \times \mathbf{B} = 0 \quad \checkmark$$

$$\text{e. } \mathbf{E} = \underset{A_r}{\hat{r}} \left(\underbrace{3 - \frac{r}{1+r}}_{A_r} \right) + \underset{A_z}{\hat{z}}z$$

$$\nabla \times \mathbf{E} = 0 \quad \checkmark$$

A field is conservative when each component depends only on the coordinate associated with that component (e.g. $\hat{x}$ component depends only on x , $\hat{y}$ component depends only on y , and $\hat{z}$ component depends only on z).

Summary



	in	out	Calculation
$\text{grad } f$ ∇f	scaler	vector -	$\nabla f = 0$ everywhere uniform
$\nabla \cdot \vec{A}$ div	vector	scaler	$\nabla \cdot \vec{A} = 0$ everywhere Solenoidal
$\nabla \times \vec{A}$ curl	vector	vector	$\nabla \times \vec{A} = 0$ everywhere conservative



given a scalar field $f \Rightarrow \nabla f \rightarrow \nabla \times (\nabla f)$

∇f is always a conservative field

$$\nabla \cdot (\nabla f) = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2}$$

Laplacian operator $\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$
"del square"

expand

Laplacian of a vector:

given $\vec{A} = \hat{x}A_x + \hat{y}A_y + \hat{z}A_z$

$$\nabla^2 \vec{A} = \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) \vec{A} = \hat{x} \nabla^2 A_x + \hat{y} \nabla^2 A_y + \hat{z} \nabla^2 A_z$$

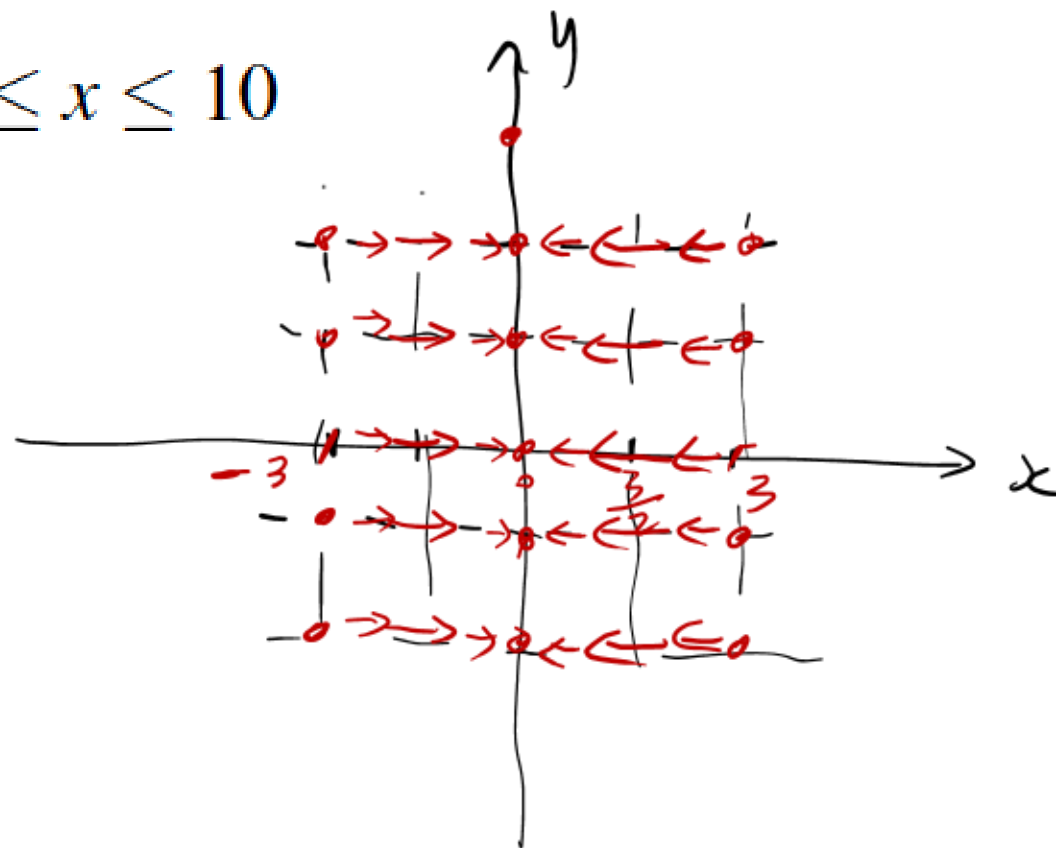
$$\begin{aligned} \nabla f &= \hat{x} \frac{\partial f}{\partial x} + \hat{y} \frac{\partial f}{\partial y} + \hat{z} \frac{\partial f}{\partial z} \\ \nabla \times (\nabla f) &= \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \frac{\partial f}{\partial x} & \frac{\partial f}{\partial y} & \frac{\partial f}{\partial z} \end{vmatrix} \\ &= \hat{x} \left(\frac{\partial}{\partial y} \frac{\partial f}{\partial z} - \frac{\partial}{\partial z} \frac{\partial f}{\partial y} \right) + \hat{y} \left(\frac{\partial}{\partial z} \frac{\partial f}{\partial x} - \frac{\partial}{\partial x} \frac{\partial f}{\partial z} \right) + \hat{z} \left(\frac{\partial}{\partial x} \frac{\partial f}{\partial y} - \frac{\partial}{\partial y} \frac{\partial f}{\partial x} \right) \\ &= 0 \end{aligned}$$

3.37 For each of the following scalar fields, obtain an analytical solution for ∇T and generate a corresponding arrow representation.

* (g) $T = x + \cos(\pi x/3)$, for $-10 \leq x \leq 10$

$$\begin{aligned}\nabla T &= \hat{x} \frac{\partial T}{\partial x} + \hat{y} \frac{\partial T}{\partial y} + \hat{z} \frac{\partial T}{\partial z} \\ &= \hat{x} \left(-\sin\left(\frac{\pi x}{3}\right) \cdot \frac{\pi}{3} \right) \\ &= -\hat{x} \sin\left(\frac{\pi x}{3}\right) \frac{\pi}{3}\end{aligned}$$

$$\begin{cases} x=0 & \nabla T = 0 \\ x=\frac{3}{2} & \nabla T = -\frac{\pi}{3} \hat{x} \quad \text{max mag.} \\ x=3 & \nabla T = 0 \end{cases}$$



$$\cos\left(\frac{\pi}{6}\right) = \frac{\sqrt{3}}{2}$$
$$\sin\left(\frac{\pi}{6}\right) = \frac{1}{2}$$

when $\phi = 0$
 $\phi = \frac{\pi}{2}$
 $\phi = \frac{\pi}{2}$

fields, obtain an an-
corresponding arrow

$\phi = \pi$

$\nabla \gamma = \hat{x}$

$\phi = \frac{\pi}{2}$

$\phi = \frac{\pi}{3}$

$\phi = \frac{\pi}{6}$

$\phi = 0$

$\phi = \pi$

$\cos \phi \sin \phi$

ϕ

$$\nabla T = \hat{r} \cos\phi - \hat{\phi} z \cos\phi \sin\phi$$

$$= \hat{r} \cos^2 \phi + \hat{\phi} \frac{1}{r} \cdot 2 \cos \phi (-\sin \phi)$$

$$\nabla V = \hat{\mathbf{r}} \frac{\partial V}{\partial r} + \hat{\boldsymbol{\phi}} \frac{1}{r} \frac{\partial V}{\partial \phi} + \hat{\mathbf{z}} \frac{\partial V}{\partial z}$$

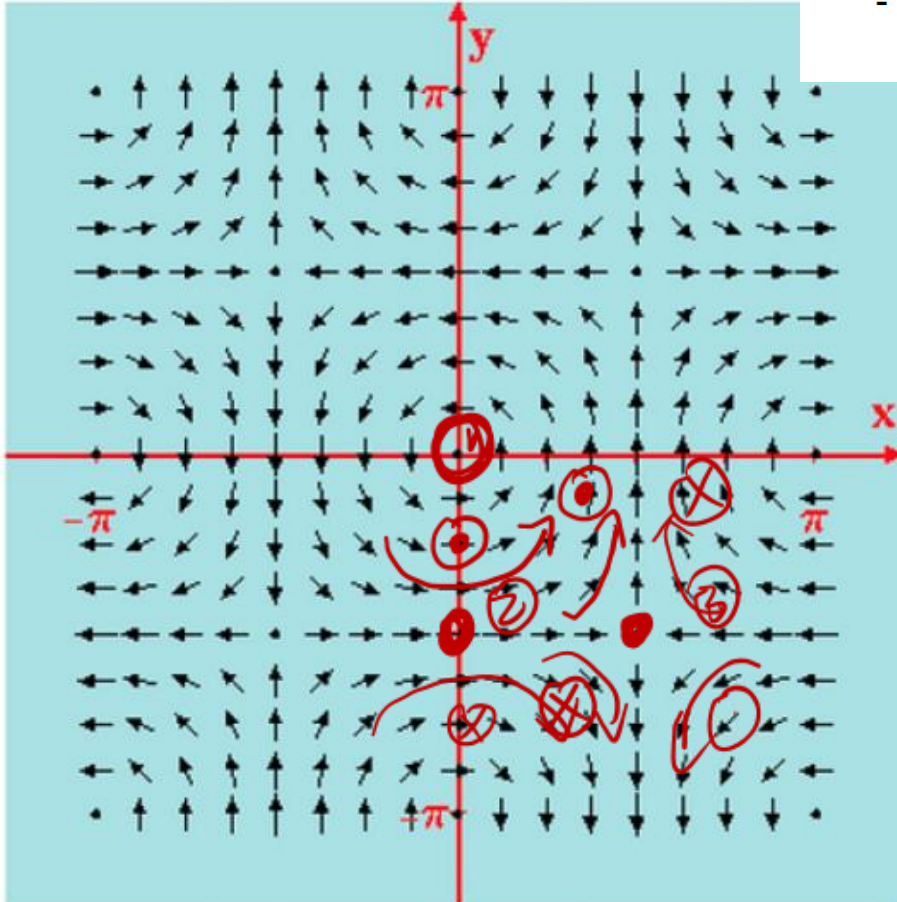
Repeat homework problem 3.44 a, b, d, e, g, and h (below), but:

In-class example

- FIRST, just look (no math) at the vector map and answer these questions:
 - Is the field solenoidal (divergenceless)? If not, where is it positive and where is it negative?
 - Is the field conservative (irrotational or no curl everywhere)? If not, where does it point in $+\hat{z}$ and where does it point in $-\hat{z}$?
 - It may help to make sketches of the maps and indicate where you expect the field to be solenoidal, conservative, or have positive or negative divergence or curl.
- SECOND, check a few of your answers (as you have time) by doing the math to find the divergence and curl.

a. $\mathbf{A} = -\hat{x} \cos x \sin y + \hat{y} \sin x \cos y$, for $-\pi \leq x, y \leq \pi$

Figure P3.44(a)



Solenoidal
non-conservative

① $\nabla \cdot \vec{A}$
 $\nabla \times \vec{A} \neq 0$

② $\nabla \cdot \vec{A} = 0$
 $\nabla \times \vec{A} = 0$

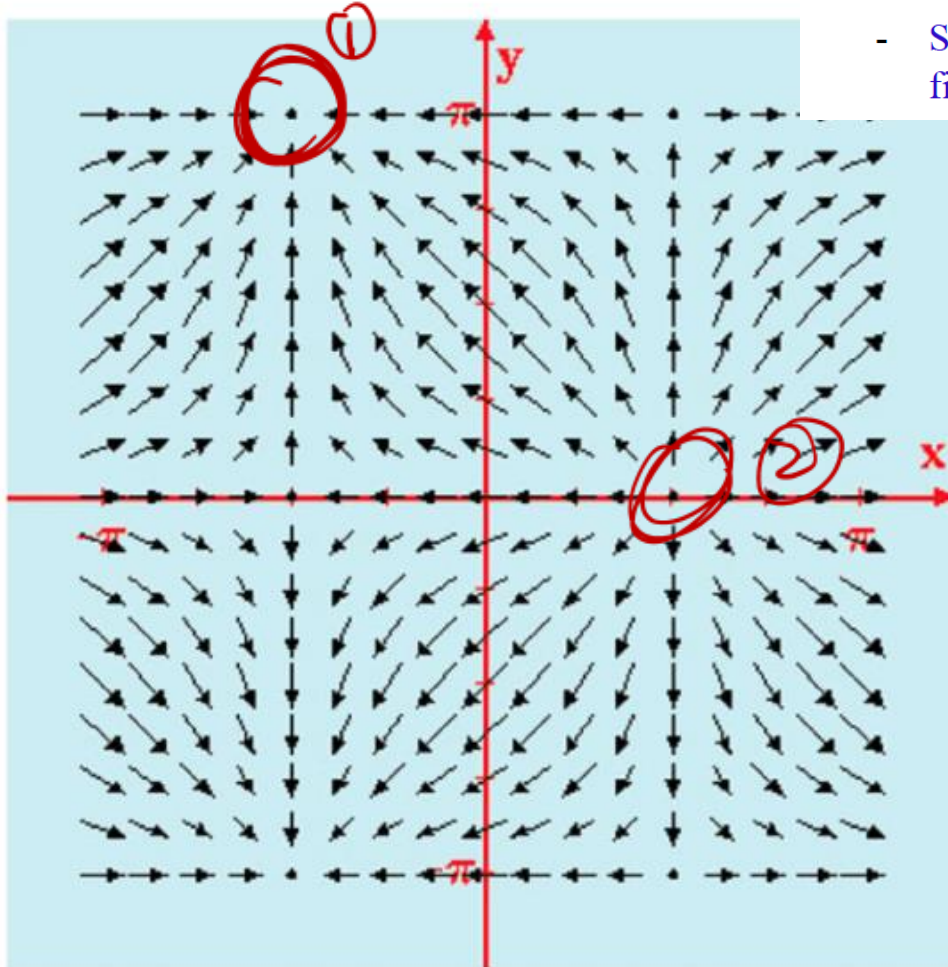
③ $\nabla \cdot \vec{A} = 0$
 $\nabla \times \vec{A} = 0$

Repeat homework problem 3.44 a, b, d, e, g, and h (below), but:

In-class example

d. $\mathbf{A} = -\hat{\mathbf{x}} \cos x + \hat{\mathbf{y}} \sin y$, for $-\pi \leq x, y \leq \pi$

Figure P3.44(d)



$\nabla \times \vec{A} = 0$

conservative field

non-solenoidal field

- FIRST, just look (no math) at the vector map and answer these questions:
 - Is the field solenoidal (divergenceless)? If not, where is it positive and where is it negative?
 - Is the field conservative (irrotational or no curl everywhere)? If not, where does it point in $+\hat{z}$ and where does it point in $-\hat{z}$?
 - It may help to make sketches of the maps and indicate where you expect the field to be solenoidal, conservative, or have positive or negative divergence or curl.
- SECOND, check a few of your answers (as you have time) by doing the math to find the divergence and curl.

① $\nabla \cdot \vec{A} < 0$ sink -

② $\nabla \cdot \vec{A} > 0$ source